Chapter 26

Securitization, Structured Credit, and the 2007–2009 Crisis

Section 24.13 established that correlation redistributes a portfolio's loss without changing it, and Section 25.7 put numbers on the redistribution: a senior tranche valued at a correlation of 0.10 and facing 0.50 has an expected loss forty-three times larger. This chapter is about what happened when an industry was built on a number no one could observe.

Every mechanism in this book that produced a loss elsewhere appears here at once: the model risk of Chapter 21, the crowding of Chapter 22, the funding fragility of Chapter 8, the rating-trigger effects of Section 24.10, and the gap between a contract and the risk it was meant to transfer from Section 25.13. It remains the best available case study because it is the only episode in which all of them fired together.

The chapter runs from the mechanism to the history and back to the present. Sections 26.1 and 26.2 are the structure: what pooling and ranking do, and why a tranche is a call spread on the pool's loss, which makes it convex in every input. Section 26.3 is the counter-example: the agency mortgage market, larger than everything that failed, where the option being sold was known and priced. Sections 26.4 to 26.6 are how the deals were built and modeled: the three sources of credit support and why they shrink when they are needed, re-securitization, and the Gaussian copula and what it was asked to do. Sections 26.7 to 26.12 are the history, ending with the order in which things broke. Sections 26.13 to 26.16 ask what was repaired, which structure survived, and where the same pattern operates now, and Section 26.17 records where the models came from.

26.1 WHAT SECURITIZATION DOES

Pool a set of loans, sell claims on the pool's cash flows, and rank those claims. That is the whole idea, and in its plain form it is useful.

It converts illiquid assets into tradeable ones. A mortgage cannot be sold easily; a security backed by ten thousand of them can.

It matches risk to appetite. A pension fund that cannot hold individual mortgages can hold a senior claim on a pool of them. Someone else takes the first-loss piece.

And it transfers risk out of banks, which is what the capital frameworks of Chapter 27 are designed to encourage when the transfer is genuine.

None of this is objectionable, and the market for plain mortgage-backed securities functioned through the crisis and functions now. What failed was not securitization; it was a specific set of things built on top of it, and keeping the two separate is necessary to draw any useful lesson.

26.2 THE WATERFALL

Cash from the pool is distributed by a **waterfall**: senior claims are paid first, then mezzanine, then equity. Losses arrive in the reverse order, so the equity tranche absorbs the first losses and the senior tranche only suffers once everything below it is exhausted.

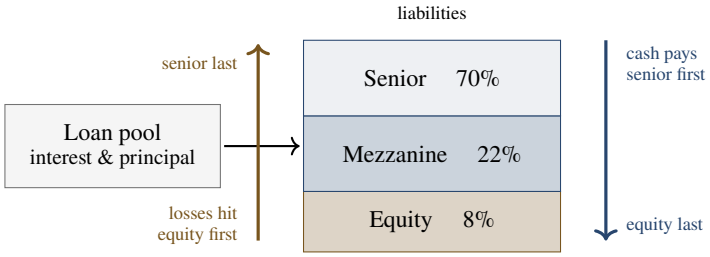


Figure 26.1: Cash descends the stack by seniority; losses climb it from the bottom. Tranche sizes illustrative.

Figure 26.1 is the entire structure, and the two arrows pointing in opposite directions are the reason a single pool can support securities with genuinely different risk. *Nothing has been created*: the pool's total cash and total loss are unchanged, and every currency unit the senior tranche is protected from is one the equity tranche absorbs.

A tranche is therefore an option on the pool's loss. Write L for the pool's loss as a fraction of its notional. The $[a, b]$ tranche absorbs

$$L_{[a,b]} = \frac{\min\big(\max(L-a,\,0),\;b-a\big)}{b-a} \tag{26.1}$$

as a fraction of its own notional: a call spread on L struck at a and b, normalized, which is Section 12.2's structure applied to a loss rather than a price. The attachment point a is the strike, the detachment point b the cap, and every property of a tranche in this chapter follows from that.

The family is named after its collateral, and the names are worth fixing because the rest of the chapter uses them. A tranched pool of mortgages is a mortgage-backed security; a tranched pool of bonds or loans is a collateralized debt obligation (CDO); when the collateral is leveraged corporate loans it is a **collateralized loan obligation** (CLO), the member of the family that survived, which Section 26.14 treats on its own terms. A CDO whose collateral is itself tranches of other deals is a **CDO-squared**, and Section 26.5 is about what that does. *Nothing in the mathematics distinguishes them*: (26.1) is the same function in every case, which is precisely why the differences that matter are in the collateral and in who manages it.

Two immediate consequences follow, and the rest of the chapter rests on them. *Adding up the tranches returns the pool*: $\sum(b_j - a_j)L_{[a_j,b_j]} = L$ when the layers partition $[0, 1]$, so securitization redistributes loss and creates none. And a tranche's expected loss is nonlinear in the pool's, because (26.1) is kinked at both ends, which is where Example 26.1's leverage comes from.

Identifying a tranche as a call spread does most of the analytical work in this chapter, because it means everything Chapter 12 said about spreads applies. A senior tranche is a deeply out-of-the-money option: worth almost nothing in expectation, enormously sensitive to the assumed distribution, and, critically, *convex in the parameters*.

Figure 26.2 shows both halves of the argument: the call-spread payoff on the left, and on the right what that payoff does to an error in the input.

Example 26.1 What a tranche does to a parameter error

A pool of 100 loans, each with a 5% default probability and 60% loss given default, correlation 0.15. The pool's expected loss is 3.00%, the 12–30% senior tranche's 0.266%.

Now suppose the true default probability was higher than assumed:

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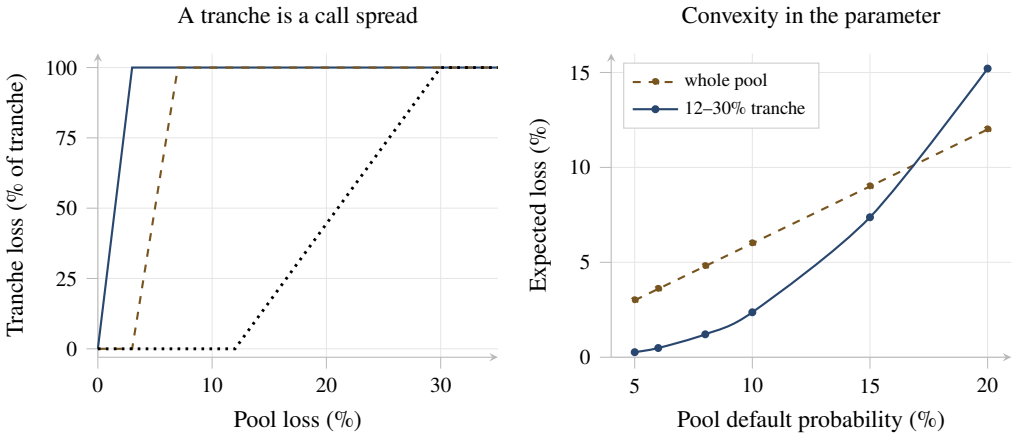


Figure 26.2: Left: equity (0–3%), mezzanine (3–7%) and senior (12–30%) losses against pool loss. Right: expected loss against default probability.

True default probability	Pool expected loss	Senior tranche expected loss
5%	3.00%	0.266%
6%	3.60%	0.486%
8%	4.80%	1.208%
10%	6.00%	2.369%
15%	9.00%	7.374%
20%	12.00%	15.205%

The numbers come from the copula exactly as Example 24.8 applied it. The tranche is untouched until 21 of the 100 loans default, so the pool's loss moves with the default probability and the tranche's with the probability of reaching that count.

Tripling the default probability triples the pool's loss and multiplies the senior tranche's by twenty-eight.

This is the chapter's central mechanism and it requires no story about fraud or stupidity. A tranche is an option, options are convex, and convexity amplifies parameter error.

The same convexity works in the parameter Section 25.7 identified. Both the default probability and the correlation enter a senior tranche's value with high leverage, and neither was observable. Two unobservable inputs, each amplified by a factor of tens, is not a model that should be used to rate an instrument AAA.

26.3 THE SECURITIZATION THAT WORKS

The counter-example comes first, because it is larger than the failures and it did not fail.

Agency mortgage-backed securities are issued or guaranteed by Ginnie Mae, Fannie Mae or Freddie Mac, and most are **pass-throughs**: the pool's monthly interest and principal, prepayments included, are passed straight through to investors in proportion to their holdings. They carry no meaningful credit risk: the guarantor absorbs default, and the investor receives principal and interest regardless. What the investor bears instead is prepayment risk, and it is a pure option problem with no correlation parameter in it.

A borrower may repay at any time without penalty. So the investor in a mortgage pool is short a call on the loan, exercised when refinancing is attractive, which is when rates have fallen.

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Example 26.2 Negative convexity, measured

A 30-year 5.5% pass-through on a pool of level-payment mortgages (a constant monthly installment over the loan's life), paying monthly, at a 5% yield compounded monthly. The speed is the market's conditional prepayment rate (CPR), the fraction of the outstanding balance repaid early in a year beyond scheduled amortization, applied monthly as $1-(1-\text{CPR})^{1/12}$ of the balance left after scheduled principal; it responds to the refinancing incentive: slow while market rates sit above the coupon, steep once they fall below it, leveling off once those who can refinance have. Prices are per unit of face at 5% + Δy; "fixed speed" keeps the 12% base rate; the comparison is a seven-year 5.5% semiannual bullet, a bond repaying all its principal at maturity, seven years being about the pool's average life at the base speed (6.6 years), so the comparison isolates the option rather than maturity.

Yield change	CPR	Pass-through price	Price at fixed 12% CPR	7y bullet
−2.0%	35%	1.0534	1.1399	1.1568
−1.0%	22%	1.0496	1.0797	1.0908
0	12%	1.0253	1.0253	1.0292
+1.0%	8%	0.9698	0.9759	0.9718
+2.0%	6%	0.9035	0.9307	0.9181

	Effective duration	Effective convexity
Pass-through, prepayments respond	3.89	**−304**
Same bond, fixed prepayment speed	5.07	+48.5
Seven-year bullet	5.78	+39.9

Effective duration and convexity are read off the ±1% prices: for the responding pass-through, $(1.0496-0.9698)/(2\times1.0253\times0.01) = 3.89$ and $(1.0496+0.9698-2\times1.0253)/(1.0253\times0.01^2) = -304$.

The convexity is negative, and strongly so. A two-percent rally lifts the bullet by 12.4% and the pass-through by 2.7%; a two-percent sell-off costs the bullet 10.8% and the pass-through 11.9%. The investor gets a fraction of the upside and all of the downside, because the borrowers refinance when rates fall and stay put when they rise.

The mechanism is exactly Chapter 18's. The investor is short an option, so they are short gamma, and Section 18.12's account of what short gamma does to a hedger applies directly: as rates fall the pool's duration shortens, so a mortgage portfolio hedged to a duration target requires buying duration as rates fall and selling it as they rise. That is mechanical buying into a rally and selling into a sell-off, pushing rates further in the direction they have already moved, by an investor base measured in trillions, and it is one of the largest identifiable feedback flows in any fixed-income market.

Two lessons follow for this chapter.

Risk that is genuinely understood can be borne at enormous scale. The agency mortgage-backed securities market is bigger than everything that failed in 2008 put together, its central risk is harder to model than credit default, and it worked. The difference is that the option being sold was known to be an option, priced as one, and hedged as one.

And the failure of Section 26.5 was not about complexity. Prepayment modeling is more technically demanding than a Gaussian copula. What distinguishes the two is not difficulty but whether the model's key input could be checked against anything, and prepayment speeds are observed monthly while default correlation was observed never.

26.4 HOW A DEAL IS ACTUALLY BUILT

A bond's **credit support** is the loss the pool absorbs before the bond takes any, and it comes from three sources. **Subordination** is Section 26.2's tranching: for a senior bond attaching at 20%, the 20% of bonds beneath it. **Overcollateralization** is the excess of assets over liabilities:

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$980 million of bonds against $1,000 million of loans gives 2% of cushion. **Excess spread** is what is left of the pool's coupon after the bonds' coupon and the servicing fees are paid; it is the first thing losses consume, before any tranche is touched, and if it is retained rather than released to the equity it accumulates, so the support grows every year the pool performs.

This is why the structures looked so robust, and the flaw is visible once stated. The support grows *provided the excess spread is actually earned*, and it is earned only while the loans keep paying. A pool that starts defaulting loses its spread at the moment it begins consuming the cushion it already has. This correlation between the buffer and the loss is structurally the wrong-way risk of Section 9.9 and the recovery–default correlation of Section 24.4: a protection that shrinks in the state where it is needed. Example 26.3 runs the two paths side by side.

Example 26.3 Where the protection comes from

A $1,000 million pool of loans with an 8.5% weighted-average coupon backs $980 million of bonds costing 5.5%; servicing and fees take 0.5%, and every percentage below is of the original $1,000 million. The excess spread is $8.5 - 5.5 - 0.5 =$ **2.5%** a year on the paying balance, $25 million in a default-free year, and the overcollateralization is 2%.

The promised path. A senior bond attaching at 20% starts with 20% + 2% = 22% of support. With the excess spread retained, each year adds 2.5%: 24.5% at the end of year 1, then 27.0%, 29.5%, 32.0% and 34.5%, the table's "promised" column.

The stressed path. Suppose 8% of the original balance defaults each year at a 40% recovery. The pool then loses $8\% \times 60\% = 4.8\%$ a year, but its performing balance falls by the full 8%: a defaulted loan pays no interest, and what is recovered on it comes back as principal, not as a loan still earning the coupon. Excess spread is earned on the average performing balance of the year, and losses are absorbed by the overcollateralization and the spread earned before they touch subordination:

	Promised	Realized under stress			
Year	Support	Spread earned	Loss	OC left	Support
1	24.5%	2.40%	4.80%	0.00%	19.60%
2	27.0%	2.20%	4.80%	0.00%	17.00%
3	29.5%	2.00%	4.80%	0.00%	14.20%
4	32.0%	1.80%	4.80%	0.00%	11.20%
5	**34.5%**	1.60%	4.80%	0.00%	**8.00%**

Years 1 and 2 are worth working in full. In year 1 the performing balance falls from 100% to 92%, averaging 96%, so the deal earns $2.5\% \times 0.96 = 2.40\%$ of spread against 4.80% of loss. The 2% of overcollateralization and the 2.40% together absorb 4.40%, so the overcollateralization is gone and the remaining 0.40% comes out of subordination: support ends the year at $22.0 + 2.40 - 4.80 = 19.60\%$. In year 2 the balance falls from 92% to 84%, averaging 88%, earning $2.5\% \times 0.88 = 2.20\%$ against another 4.80%; with the overcollateralization gone, the whole 2.60% shortfall comes out of subordination, and support falls to $19.60 - 2.60 = 17.00\%$.

The support does not rise to 34.5%; it falls to 8.00%, a swing of 26.5 points, and through two channels at once: the losses consume it, and the shrinking performing balance earns less of it each year, 1.60% in year five against 2.40% in year one. Cumulative losses of 24.0% have consumed the overcollateralization, all ten points of spread earned, and twelve of the twenty points of subordination; at more than three points a year, two and a half more years exhaust it. *The senior bond is on course not to survive.*

Figure 26.3 draws this table beside Example 26.2's: on the left the pass-through's price bending the wrong way, on the right the two support paths parting from the first year. Nothing in the example requires a copula, a correlation, or a view about house prices: it requires only that defaults arrive and that the spread be earned on what is left. The models that projected

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the accumulating column ran only scenarios in which it accumulated. The protection was a projection presented in a column that looked like a fact.

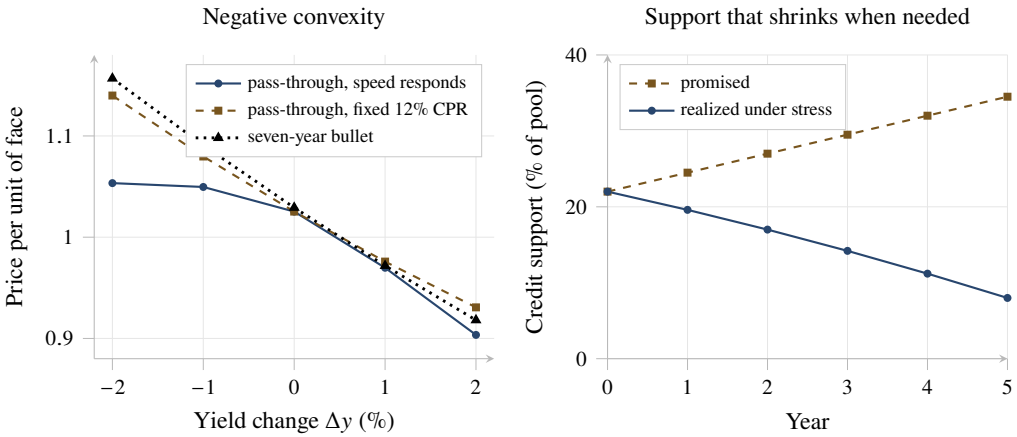


Figure 26.3: Left: the prices of Example 26.2. Right: the promised and stressed support paths of Example 26.3.

26.5 RE-SECURITIZATION

The step that turned a bad decade into a crisis was doing it twice.

Mezzanine tranches are hard to sell: too risky for senior buyers, not levered enough for equity buyers. The solution the market found was to pool them and tranche the pool.

Example 26.4 A CDO of mezzanine tranches

Take the pool of Example 26.1. At a correlation of 0.15 its tranches have expected losses:

Equity 0–3%	64.07%
Mezzanine 3–7%	20.19%
Mezzanine 7–12%	4.45%
Senior 12–30%	0.27%
Super senior 30–100%	0.00%

Now take the 3–7% mezzanine piece from one hundred *different* deals and pool those. Each has an expected loss of 20.19%. Tranche the new pool, and everything depends on the correlation *between* the mezzanine pieces. Model it as the copula applied twice, in Section 24.13's notation. Loan j of deal i has $X_{ij} = \sqrt{0.15}\,M_i + \sqrt{0.85}\,Z_{ij}$, so the loans in a deal share that deal's factor M_i at the correlation 0.15 of Example 26.1; and the hundred deal factors themselves load on one market factor M, $M_i = \sqrt{\rho}\,M + \sqrt{1-\rho}\,Z_i$, where ρ is now the inter-deal correlation. Conditional on M the hundred pieces are independent, each with the loss distribution its own tranche loss function gives it, so the new pool's loss distribution is built exactly as in Section 25.7's Step 2, with a piece's loss distribution in the place a single default occupied. The four layers of the re-securitization then lose:

Inter-deal ρ	"AAA" 30–100%	"AA" 12–30%	3–12%	Equity 0–3%
0.10	1.08%	44.13%	94.36%	99.96%
0.30	3.83%	40.89%	80.19%	97.71%
0.60	7.55%	36.64%	63.34%	86.84%
0.90	11.25%	32.18%	49.18%	69.72%

Weight the columns by width (0.70, 0.18, 0.09, 0.03) and every row adds to 20.19%, the pool's

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expected loss, as it must. The senior tranche of the re-securitization goes from losing 1.08% of its notional to 11.25% as the inter-deal correlation rises, a factor of ten, driven entirely by a parameter about which no market quoted anything.

And the parameter was not 0.10. The mezzanine tranches of one hundred subprime deals from the same vintage are exposed to one thing: national house prices. They are not a hundred independent bets on a hundred pools; they are one bet, made a hundred times, on whether house prices keep rising.

So the diversification that justified the structure did not exist. Pooling a hundred claims removes idiosyncratic risk only if the risk is idiosyncratic, and the claims being pooled had already had their idiosyncratic risk stripped out by the first securitization, which is precisely what a mezzanine tranche is. *Re-securitizing tranches pools the systematic residual, and the systematic residual does not diversify.*

26.6 THE COPULA AND WHAT IT WAS ASKED TO DO

Section 24.13's one-factor Gaussian copula was the industry standard, and it deserves a more careful verdict than it usually receives.

As a device it is unobjectionable. It is a way of writing down a joint distribution with a single correlation parameter, and it makes the loss distribution computable, which is a legitimate reason to choose a model. Section 25.7 noted that the market's own use of it, quoting tranches by the implied correlation that reproduces their prices, was self-evidently a quoting convention rather than a belief: a single-parameter model producing different parameters at different attachment points is announcing its own misspecification, exactly as the volatility smile does in Chapter 19.

Combining (26.1) with that factor model gives every tranche number in this chapter and the last, and the calculation is Steps 2 and 3 of Section 25.7: conditional on the common factor the defaults are binomial with the probability $p(M)$ of equation (24.12), and one integral over the factor and one binomial sum price the whole capital structure, which is why the model was adopted and could be run on every deal. As the pool grows, its loss fraction becomes $\ell\, p(M)$, a function of the factor alone, and the **large-pool** loss distribution is Chapter 24's worst-case default rate (24.15) read in reverse: with each issuer's default probability π, correlation ρ and loss given default $\ell = 1 - R$, the pool loses more than $\ell\,\mathrm{WCDR}(q)$ with probability exactly $1 - q$.

That is the formula underneath the Basel capital charge and, read backward, the reason a senior tranche is a statement about ρ. As $\rho \to 0$ the distribution collapses onto $\ell\pi$ and a senior tranche can never be touched; as $\rho \to 1$ it becomes a coin flip between nothing and everything. Rating such a tranche is choosing a point between those two, which is why the letter cannot be read without ρ.

Figure 24.6 drew the matching distribution of defaults for Example 26.1's pool. What is visible there is the middle draining into both tails as ρ rises, the mean pinned at five defaults, a loss of $\ell\pi = 3\%$; what the senior tranche cares about is the part the figure can barely show (the mass beyond the 12% attachment point, 21 defaults, at the right edge of its axis), which the large-pool approximation puts at 1.24% at $\rho = 0.15$, against the 1.50% that Example 26.1's exact count of a hundred loans gives. That mass is the senior tranche. Figure 26.4 draws the probability that the pool's loss exceeds x on the logarithmic scale the tail needs: left of the 12% line the loss is absorbed by the layers beneath; the height of each curve at the line is the probability the senior tranche is touched at all. The curves are nearly straight because this tail is close to exponential in x; what ρ sets is its slope.

The loss distribution is not yet a price. Section 25.7 gave the four steps from one to the other and Example 25.5 ran them on a 125-issuer index portfolio, so the calculation is not

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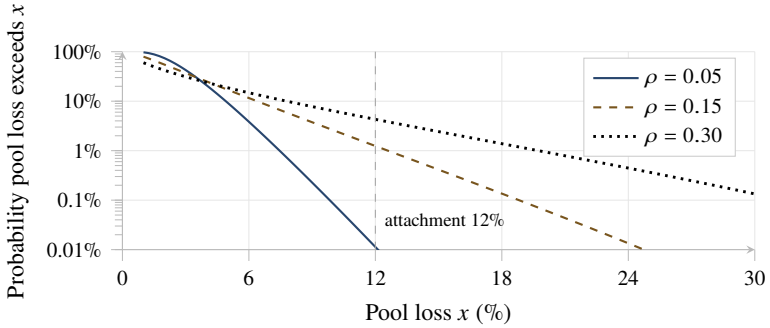


Figure 26.4: Probability that the pool's loss exceeds x, from the large-pool limit of equation (24.14), inverted as in equation (24.15), with $\pi = 5\%$, $\ell = 60\%$, evaluated every 0.25%; logarithmic scale.

repeated; only the principle is restated. The risky annuity is the present value of the notional that survives and the protection leg of the notional that does not, and where a CDS weighted by $Q(t_i)$, a tranche weights by $1 - E_{[a,b]}(t_i)$, the expected fraction still outstanding. Change the product and only the definition of "survives" changes (an issuer's default, a layer's erosion, a pool's amortization for Section 26.4's bonds), which is why the CLO of Section 26.14, whose cash flows are governed by tests rather than by a copula, is priced by projecting the same two quantities through its waterfall, not by a new theory.

Three things went wrong, and only one of them is a criticism of the model.

The Gaussian copula has no tail dependence. **Tail dependence** is the tendency of extreme outcomes to arrive together: the probability that one variable is extreme given that another is, as "extreme" is pushed ever further into the tail. Under the Gaussian copula, extreme events are asymptotically independent: conditional on one issuer defaulting catastrophically, the probability that another does converges to its unconditional value. *This is empirically false for credit*, where defaults cluster in exactly the states the model treats as independent, and it biases senior tranches, the ones that only matter in those states, in the direction of looking safe.

The correlation was calibrated to a period that contained no test. Whatever number was used came from data in which US house prices had not fallen nationally, so the estimate could not contain the event that mattered. This is not a modeling error at all; it is the sampling problem of Section 19.9's pre-1987 discussion, and no choice of copula would have repaired it.

And the output was treated as a measurement. A model that turns an assumption into a rating, run by an institution the issuer pays, produces a number whose precision is inherited from an input no one checked. This is the failure, and it is institutional, not mathematical.

The verdict is that the copula was a bad model used badly, and that the second clause carries most of the weight. A better model calibrated to the same data by the same institutions with the same incentives would have produced approximately the same ratings.

26.7 THE WARNING NO ONE READ: MAY 2005

Two years before the crisis, the correlation market produced a clean, self-contained failure that demonstrated the model's defect precisely. It is worth recounting because everything that mattered in 2007 is visible in it, at a scale small enough that no one was harmed much.

The standard trade was to be long the equity tranche of an index and short the mezzanine, in a ratio that made the package neutral to the index spread. Section 25.7's table explains the position: the equity tranche is long correlation and the mezzanine is much less so, so the package was a way to be long correlation while hedged against the credit market's direction. It carried positively, it had performed for years, and it was widely held.

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In May 2005 two large US automakers were downgraded to junk.

The shock was idiosyncratic, and the model had no room for idiosyncratic shocks. In the one-factor copula of Section 24.13, an issuer defaults because the common factor moves or because its own draw is bad, and the second channel is a residual that diversifies away in a 125-issuer index. A downgrade large enough to move two issuers' spreads violently without moving the other 123 is, in the model's terms, an event of negligible probability.

The effect on the two legs went the wrong way on both.

The equity tranche fell, because idiosyncratic widening in a few issuers is exactly what an equity tranche is exposed to. *The mezzanine did not widen to compensate*, because the hedge ratio had been computed on a model in which spreads move together: the short mezzanine leg was sized to gain from a widening that never came, and as implied correlation fell the mezzanine spread tightened instead.

The trade lost money on both legs simultaneously, which the model said could not happen. Implied correlation moved in ways no single parameter could reconcile, and the base-correlation apparatus of Section 25.7 was in part a response to the resulting inconsistencies.

Three lessons were available in 2005 and were drawn in 2008 instead.

The hedge ratio was model output, not market data. A position hedged by a model is hedged against the model's world, and Section 21.11's measurement of model risk is the same observation with a barrier option in place of a tranche.

A parameter that must be inferred from the prices it explains is not a parameter. The market was calibrating ρ to tranche prices and then using ρ to hedge tranches, which is circular in a way that produces sensible numbers right up until the relationship it assumes fails.

And crowding turned a model error into a market event. The trade was widely held in the same direction, so the unwind moved the prices being unwound—the mechanism of Chapters 22 and 23 in a market where no one had yet thought to look for it.

The episode was small, contained, and correctly diagnosed at the time. What did not happen is the transfer of that diagnosis to the much larger book of the same model being used, on much less observable portfolios, to rate instruments AAA.

26.8 RATINGS: THE SAME LETTER, DIFFERENT RISK

Section 24.10 said the rating on a structured tranche depended on a correlation assumption the letter did not disclose. Here is what that meant in practice.

A AAA corporate bond and a AAA senior tranche were notionally calibrated to the same idealized default probability; their loss distributions in bad states were entirely different.

A corporate bond's rating can be wrong by one notch. If the issuer deteriorates it becomes AA, then A, and a holder can sell into a market that reprices gradually.

A tranche's rating is closer to binary. Example 26.1 took a senior tranche from 0.27% to 15.2% expected loss on a default-probability move that raised the pool's own expected loss only fourfold, from 3% to 12%. There is no gradual path; the tranche is fine until the attachment point is threatened and then it is not.

Two structural features made this worse.

Rating triggers were embedded in contracts. Downgrades forced sales by mandate-constrained holders and triggered collateral calls, which is Section 24.10's mechanism operating on an instrument whose downgrades, when they came, came in blocks of many notches at once.

And the ratings were used as a substitute for analysis. An investor holding a AAA tranche was, in practice, holding an agency's correlation assumption. The rating was not a summary of an analysis the investor could have done themselves; it was the entire analysis, which is a different thing and was not disclosed as such.

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26.9 THE INCENTIVE CHAIN

The technical story above is sufficient to produce large losses. It is not sufficient to produce a crisis, and the additional ingredient was that no one in the chain retained the risk they created.

A mortgage was originated by a broker paid a fee per loan, sold to a bank that pooled it, tranched by an arranger paid a fee per deal, rated by an agency the arranger paid, and sold to an investor who trusted the rating. At every step the pay was for volume and the risk moved on.

Originate-to-distribute is not inherently wrong. It is the same separation of origination from risk-bearing that makes any capital market work, and Section 24.16's discussion of banks distributing credit describes the same thing functioning correctly.

What made it fail was that the distribution was not real. Banks retained **super-senior** tranches (the top layer of the capital structure, above even the AAA-rated senior notes) because they were unsellable and appeared riskless, held them in off-balance-sheet vehicles funded with short-term commercial paper, and provided liquidity backstops to those vehicles. *The risk did a full circle and came back*, and the capital framework of the time did not capture it because the exposure was contingent and the vehicle was not consolidated.

That is the mechanism Chapter 27 exists to address, and Section 26.13 records the response.

26.10 THE EUROPEAN ALTERNATIVE

Europe funded mortgages differently, and the contrast isolates the feature that did the damage.

A **covered bond** is a bank's bond secured on a pool of mortgages that *stays on its balance sheet*. The holder has **dual recourse**: a claim on the bank and, if it fails, a preferential claim on the ring-fenced pool. The pool is dynamic: the issuer must replace loans that deteriorate.

No risk is transferred at all, which is exactly the point.

Covered bonds have a very long history and a default record close to nil, through the crisis included. The reasons map one-to-one onto the failures of this chapter.

The originator keeps the credit risk, so Section 26.9's chain has no one to pass it to. An issuer that writes bad loans must replace them out of its own assets.

There is no tranching, so Example 26.1's convexity cannot arise.

And there is no re-securitization, because there are no tranches to re-pool.

The trade-off is equally clear and should be stated. *Covered bonds do not free up bank capital*, because the risk never leaves. They are a funding instrument, not a risk-transfer instrument, and a banking system financed entirely this way would lend less per unit of capital.

So the comparison is not that Europe was wiser. It is that two systems made different choices about whether risk transfer was the objective, and the system that did not attempt it did not suffer the failures that attempting it produced. Securitization's benefits and its failure modes come from the same feature, and a reader deciding what to think about the instrument should hold both halves of that sentence at once.

26.11 SYNTHETIC CDOS AND THE SUPPLY PROBLEM

Section 25.15 established the distinction. A cash CDO buys the assets; a synthetic one sells protection on a reference portfolio via the CDS of Chapter 25.

The synthetic format removed the constraint that there had to be a loan. Demand for AAA-rated yield exceeded the supply of mortgages, and a synthetic structure could reference the same pool as many times as there were counterparties willing to take the other side.

This has two consequences, and they should be separated because they are often conflated.

It multiplied the exposure without multiplying the risk of default. A hundred synthetic deals referencing the same mortgages do not make them more likely to default; they create a hundred times the notional of contracts whose value depends on whether they do.

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And it therefore multiplied the interconnection. The loss on the mortgages was bounded; the transfer of that loss through the system was not. A bounded shock propagated through an unbounded web of contracts is a different problem from a large shock, and it is why the crisis's transmission was through counterparty exposure rather than through mortgage losses alone.

A fair counter-argument runs as follows. Synthetic structures also let the risk be held by whoever wanted it, including investors who correctly thought subprime overvalued and used the format to be short. The instrument was symmetric; the understanding was not.

26.12 WHAT ACTUALLY FAILED

Case Study 26.1 The order in which things broke

The sequence matters, because the popular account compresses it and loses the mechanism.

2006–2007: the pool assumptions were wrong. Subprime delinquencies rose above anything in the calibration sample, for reasons (loosening underwriting, adjustable-rate resets, house prices flattening) visible in the data before they reached the ratings.

Mid-2007: tranches repriced nonlinearly. Example 26.1's convexity worked: bad pool losses became catastrophic tranche losses, mezzanine and re-securitized tranches first. In stylized round numbers, not the record of any one pool: if falling house prices double cumulative defaults from 15% to 30% of a pool's balance, and a foreclosed loan loses half its balance, the pool's loss doubles from 7.5% to 15%. A mezzanine tranche attaching at 8% and detaching at 20% loses nothing at 7.5% and $(15-8)/12 = 58\%$ of its principal at 15%.

Late 2007: the funding withdrew. The off-balance-sheet vehicles of Section 26.9 funded long-dated tranches with short-dated paper. When the paper could not be rolled, the sponsoring banks took the assets back onto their balance sheets: Chapter 8's funding-fragility problem, arriving by a route the capital framework had not anticipated.

2008: the counterparty channel opened. Institutions that had sold protection on senior tranches faced collateral calls as the marks fell. The largest case was an insurer that had written protection without capital against the mark-to-market exposure, reasoning that the tranches would never default; they did not need to, because the collateral calls on the marks sufficed.

And then valuation itself failed. With no trades, the marks came from models, and Chapter 21's model-risk warning arrived at the worst moment: institutions marking the same instrument differently, disputing collateral calls, unable to say who was solvent. Chapter 9's XVA machinery and Chapter 8's collateral standards exist because of it.

Three observations follow for a reader who was not there.

The mortgage losses alone were survivable. Realized US subprime losses were large but below the banking system's capital. The crisis was in the transmission, not the shock.

The instruments performed as written. Senior tranches largely did pay, eventually. Holders who could fund them to maturity mostly recovered. The losses were concentrated in those who had to sell, Section 5.15's limits of arbitrage at systemic scale.

And the risk had not been transferred. That is the finding that should be carried forward from this chapter, because it is the one that generalizes: a structure that appears to move risk out of a system, while the system retains the funding, the liquidity backstop, and the counterparty exposure, has not moved the risk.

26.13 WHAT WAS FIXED

More was fixed than is usually acknowledged, and a fair account matters for Section 26.16.

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Risk retention. Sponsors must retain an economic interest (typically 5%) in what they securitize. This addresses Section 26.9 directly, and its weakness is that 5% of a structure is a small fraction of the fee income from arranging it.

Consolidation and capital. Off-balance-sheet vehicles are consolidated; liquidity backstops carry capital; re-securitizations carry punitive risk weights that have made CDO-squared structures uneconomic rather than merely unfashionable.

Rating agency reform. Methodologies must be published, ratings on structured products are labeled distinctly, and many regulatory rules no longer rely on ratings. That is the most important change, since the mechanical use of ratings is what made downgrades destructive.

And the infrastructure of Chapters 2 and 8. Central clearing, initial margin on uncleared trades, and standardized collateral terms address the counterparty channel that transmitted the crisis.

The structure that failed cannot easily recur. A CDO-squared of subprime mezzanine, rated by an issuer-paid agency, held in an unconsolidated vehicle funded with commercial paper, is blocked at each link: the re-securitization by risk weights, the vehicle by consolidation, the paper funding by the capital its backstop carries, and the rating by rules that no longer use it.

What has not been fixed is the general pattern, and that is Section 26.16.

26.14 THE STRUCTURE THAT SURVIVED

A CLO is worth setting out on its own terms rather than only by contrast, because it is the member of the family a reader is most likely to meet.

The loans come first. A manager accumulates them in a **warehouse**, a credit line from the arranging bank with the future equity investors bearing the price risk; at closing the deal issues perhaps \$400–500 million of notes from AAA down through BB with an unrated equity piece beneath, repays the warehouse, and completes the portfolio in a short ramp-up: typically 150–300 leveraged corporate borrowers, floating rate, senior secured. During a **reinvestment period** of about five years, principal received is reinvested rather than returned, subject to eligibility and diversity constraints written into the indenture. The manager trades within those constraints and usually holds part of the equity.

The feature that matters most is the one the 2007 structures lacked: a set of **coverage tests** that redirect cash automatically as the portfolio deteriorates. The senior **overcollateralization test** compares collateral par with the notes it supports,

$$\text{OC} = \frac{\text{collateral par}}{\text{par of this tranche and everything senior to it}} \tag{26.2}$$

and if it breaches its trigger, interest due to the equity amortizes the senior notes until it passes.

Example 26.5 The test that deleverages the deal automatically

A \$400 million CLO with \$256 million of AAA notes and a senior overcollateralization trigger of 120%. The opening ratio (26.2) is $400/256 = 156.25\%$.

The test fails once collateral par falls below $1.20 \times 256 = 307.2$, after \$92.8 million of erosion, or 23.2% of the pool.

Suppose it does. At \$300 million of par the ratio is 117.2%, a breach of 2.8 points of overcollateralization. Restoring it requires paying down the AAA by X where $300/(256-X) = 1.20$, so

$$X = 256 - \frac{300}{1.20} = 6.0$$

Six million dollars, one and a half percent of the deal, cures a breach that looked like 2.8 points. The diversion works on the *denominator*: paying down senior notes shrinks the base the ratio is measured against.

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At $280 million of par the ratio is 109.4% and the required pay-down is $22.7 million, so the cure gets rapidly more expensive as the collateral erodes.

Two consequences follow. The deal deleverages itself with no one deciding to, which Chapter 8's funding spirals show does not happen when deleveraging needs a decision under stress. And *the equity holder pays for it* (the diverted cash is the equity's distribution), so the party with the most information and the most incentive to gamble for resurrection is the one whose cash is taken first. Compare Example 26.1, where nothing stood between a parameter error and the senior tranche.

A CLO is structurally the same object as the instruments in Section 26.5 (a pool, a waterfall, a AAA senior piece), and CLOs are now the largest structured credit market. They performed well through 2008 and through 2020, and the reasons are specific enough to be diagnostic.

The collateral is not exposed to one factor. Loans to a hundred companies in many industries carry real idiosyncratic risk; the subprime mortgages of one vintage did not: Example 26.4's point. The diversification a CLO claims is real, and that is the largest difference.

There is no re-securitization. CLOs hold loans, not other deals' tranches; Section 26.5's compounding cannot start.

The liabilities are term-funded and match the assets. A CLO issues notes with maturities matched to the loan portfolio and cannot be forced to sell when spreads widen. Section 26.12's funding withdrawal has no analog—and this, rather than the credit quality, is probably the decisive difference. The instruments that failed in 2008 were largely fine on a hold-to-maturity basis; what killed their holders was having to sell.

And a manager has capital at risk. A CLO is managed within constraints, and the manager holds equity: Section 26.9's alignment problem addressed by construction, not regulation.

What is worth watching. Loan documentation has weakened over the cycle: **covenant-lite** loans, which carry no maintenance covenant (a financial ratio the borrower must keep meeting while the loan is outstanding), are now standard, so the early warnings a covenant breach used to provide arrive later. Recoveries on leveraged loans have been trending below historical averages, which Section 24.4's identification problem makes easy to underweight. And the investor base for the AAA tranches is concentrated.

26.15 THE LEVERAGED LOAN MARKET

The previous section described the structure and named its collateral in passing: leveraged corporate borrowers, floating rate, senior secured. That collateral is a market in its own right, and the two can no longer be understood separately: collateralized loan obligations are the largest single buyer of leveraged loans, and leveraged loans are very nearly the whole asset side of collateralized loan obligations.

The scale is the first thing to state. The US institutional leveraged loan market stood at about **$1.55 trillion** at the end of 2025 on the Morningstar LSTA index, more than three times the $465 billion it measured at the end of 2010, with Europe at more than €300 billion; collateralized loan obligations own about **two-thirds** of the US total. A market that was a bank funding channel in 2005 has therefore grown past the US high-yield bond market, which stood at about $1.48 trillion of par, and it did so while the credit derivatives market of Chapter 25 was shrinking by an order of magnitude. Any statement about where corporate credit risk is traded has to account for both movements.

Why it is a credit market and not a loan book. A leveraged loan is originated by a bank and then distributed. The arranging bank underwrites the whole facility, syndicates it to institutional investors, and retains little or none of it, so the instrument is priced by a market rather than held to maturity by its originator. It trades in a secondary market on standardized documentation, it

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is quoted daily, it settles through an assignment process, and index providers publish total-return indices on it. The economics are those of a floating-rate senior secured bond with a call at par, and the borrower is typically a company owned by a private equity sponsor, which is what supplies the leverage the name refers to.

Who is on each side, and why. The borrower is a sponsor-owned company financing an acquisition or a recapitalization, and it wants a floating-rate instrument it can prepay without penalty, because the sponsor intends to exit. The lender is a collateralized loan obligation buying assets for a portfolio it has already sold liabilities against, a loan mutual fund or exchange-traded fund offering retail investors a floating-rate credit exposure, or a separately managed account. The arranging bank wants the fee and not the asset. Each of those preferences is visible in the instrument: the prepayment option at par exists because the borrower wanted it, the floating rate exists because the largest buyer funds itself floating, and the covenant package is light because the arranger must clear the deal in a competitive market.

That last point is where the interesting arithmetic is. The previous section observed that recoveries on leveraged loans have been running below their historical averages, and the natural reading, that the borrowers have become weaker, is not the main explanation. Example 26.6 separates what changed from what did not.

Example 26.6 Why a senior claim recovers less without the borrower getting worse

Three capital structures on the same company. Enterprise value at origination $1,200 million and total debt $1,000 million in each; only the split between first-lien debt (the senior secured loan, with the first claim on the collateral) and junior debt differs.

Structure	First lien	Junior	Decline that first impairs the first lien
Traditional	400	600	66.7%
Intermediate	700	300	41.7%
Loan-only	**1,000**	**0**	**16.7%**

A first lien is impaired when enterprise value falls below its own par, so what protects it is everything beneath it, junior debt and equity alike, measured against its own size, and historically the senior claim was a small slice of the capital structure. With total debt fixed, shrinking the junior debt to zero makes the first lien the whole of the debt and moves the enterprise-value decline that first touches the first lien from **66.7% to 16.7%**.

Now apply one distress outcome to all three. Enterprise value falls to $700 million, a decline of 41.7%:

Structure	First-lien recovery	Junior recovery
Traditional	100.0%	50.0%
Intermediate	100.0%	0.0%
Loan-only	**70.0%**	—

The same enterprise outcome pays the first lien 100% or 70% according to how much debt sits beneath it, and nothing about the borrower differs between the three columns. An average first-lien recovery computed across a market that has shifted from the first row to the third will fall even if every borrower is identical and every enterprise outcome is identical.

Two further mechanisms push the loan-only figure down from there, and they compound rather than substitute.

Covenant-lite defers the event. A maintenance covenant forces a restructuring while enterprise value is still comparatively high; without one, the borrower continues until it cannot pay, and the recovery is measured on a smaller enterprise. A further 10% of erosion before the event takes the 70.0% to **63.0%**.

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A liability-management exercise moves the collateral. Loan documents that permit transfers to unrestricted subsidiaries (affiliates outside the lenders' collateral and covenants), or that allow new money to prime the existing lenders (rank ahead of them), let a distressed borrower and some of its lenders act against the rest. If 20% of enterprise value leaves the collateral package, the recovery falls to **50.4%**.

A claim can therefore be senior, secured, and first in the waterfall, and still recover half, without any of the three words having become false. Section 24.4's point about recovery rates was that they are the least reliable input in credit modeling; this is the mechanism behind that unreliability, and it is a documentation variable rather than a credit variable.

What this does to the structure above it. A collateralized loan obligation built on loan-only borrowers is holding collateral whose recovery distribution is worse and wider than the historical series its ratings were calibrated on, which is Section 26.8's problem arriving by a new route: the letter is the same and the loss behind it is not. The coverage tests of Example 26.5 still work (they are measured on par rather than on recovery, and they still divert cash automatically), but they trigger later relative to the eventual loss than they would have on a traditional capital structure. Nothing here undoes the four structural defenses the previous section set out; it sharpens one of the weaknesses that section named, the falling recoveries.

26.16 WHERE THE SAME STRUCTURE IS NOW

Two current markets share enough of the pattern to be worth examining, and the point of examining them is not to predict a crisis but to ask which of the specific failures each shares.

Significant risk transfer. Banks buy protection on the first-loss piece of their own loans, retaining the senior exposure, to reduce regulatory capital; the market has grown fast.

What is better. The bank keeps the loans and the senior tranche, so Section 26.9's incentive problem inverts: in 2007 the originator kept none of what it made, while here it keeps every loss beyond the first-loss piece it sold, and so most of the exposure to its own underwriting. Protection is usually funded, so Section 25.15's counterparty problem is largely absent, and the loans are the bank's own, which it underwrote and can monitor.

What is the same. The capital relief depends on a supervisory judgment that the transfer is genuine, and the history of this chapter is a history of transfers that appeared genuine and were not. And the buyers are concentrated: a few specialist funds, often themselves levered.

Private credit. Section 24.16 raised it and this is the natural place to finish the argument.

What is better, and it is substantial. The institutional holders are closed-end funds with locked-up capital that cannot be run on, so there is no maturity transformation of the kind that broke the vehicles in Section 26.12. The retail "semi-liquid" vehicles are the exception: they promise periodic redemptions, usually capped at 5% of net asset value a quarter, and several had to ration them in 2026. The loans are held whole, not tranched, so Example 26.1's convexity is absent.

What is worth watching. Valuations are infrequent and produced by the holder, so the marks that would signal deterioration may not exist. That is not the same as the risk being absent, and Section 26.12 records that the last time marks became unreliable was the moment the crisis became unmanageable. Leverage has entered at the fund level. And the banks that no longer make these loans often finance the funds that do, which rebuilds the exposure at one remove.

Neither is a repeat. Both differ from the 2007 structure in ways that address its failures; both make the risk's location harder to observe. The chapter's lesson is not that any structure is dangerous but that a system loses the ability to see a risk before it loses the ability to bear it.

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26.17 WHERE THIS CAME FROM

Portfolio credit went wrong step by step, each step defensible.

Vasicek's one-factor loan-portfolio model (published 2002, circulated far earlier) underlies Basel's capital formula and is sound within its assumptions. Li (2000) then applied a Gaussian copula to default times, and the tranche market was built on it. Andersen and Sidenius (2004) had shown how to relax the assumptions that mattered (random recovery and random factor loadings), and Hull and White (2004) had given a valuation without simulation. Coval, Jurek and Stafford (2009) stated this chapter's convexity argument formally, once the losses had arrived. Gorton and Metrick (2012) supplied the funding-run account Section 26.12 rests on.

SUMMARY

Securitization pools loans and ranks claims on them, making illiquid assets tradeable and matching risk to appetite. Plain mortgage-backed securities worked through the crisis and work now; what failed was what was built on top.

A tranche is a call spread on the pool's loss, so it is an option, and options are convex in their parameters: raising the pool's default probability raises the pool's expected loss in proportion and multiplies the senior tranche's many times over. Two unobservable inputs, each amplified that strongly, are not a basis for a AAA rating. Agency mortgage-backed securities carry pure prepayment risk, harder to model than a Gaussian copula, and they worked. The difference is not difficulty but whether the key input can be checked: prepayment speeds are observed monthly, default correlation never.

Re-securitization compounded it. Re-tranching mezzanine tranches produced a "AAA" piece whose expected loss depended almost entirely on the correlation between deals, which no one could measure—and the subprime mezzanines of one vintage are one bet on national house prices.

The Gaussian copula has no tail dependence, but the failure was institutional: a model calibrated to a sample with no national house-price decline, its output treated as a measurement by issuer-paid institutions. May 2005 had staged the rehearsal. The risk that was distributed came back: banks kept super-senior tranches in unconsolidated vehicles funded with paper they backstopped. Mortgage losses were survivable; the crisis was in the transmission.

Collateralized loan obligations are the same object and performed through 2008 and 2020: diversified collateral, whole loans, term funding and a manager holding equity remove four of the crisis's five mechanisms, and coverage tests deleverage the deal at the equity's expense. The loans beneath a CLO are a market of their own, and a senior secured claim there can recover far less than its history suggests without any of its three adjectives having become false, because recovery depends on how much debt sits beneath it, a documentation variable. Significant risk transfer and private credit share parts of it, and what they share is a risk whose location is harder to observe. A system loses the ability to see a risk before it loses the ability to bear it.

FURTHER READING

The Financial Crisis Inquiry Commission's report is the public account of record and its dissents are worth reading alongside it, since the disagreements are about causation, not facts. Gorton's work on the repo run treats Section 26.12's funding mechanism and corrects the impression that the crisis was mainly about mortgage losses. On the copula, Li's original paper is short and makes none of the claims its critics attribute to it. The May 2005 episode of Section 26.7 is set out as it happened in Belsham, Vause and Wells's Bank of England *Financial Stability Review* article on credit correlation, a diagnosis two years early. Coval, Jurek and Stafford's analysis of structured finance is the formal statement of Example 26.1's convexity argument. On what was fixed, the Basel documents on securitization capital are specific, and the right source for

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judging whether a current structure shares the defects. On private credit and significant risk transfer, the supervisory literature is where the structures and their risks are documented, and both markets are moving.

PRACTICE QUESTIONS

26.1 State three legitimate purposes of securitization, and explain why the failure of 2007–2009 does not condemn all of them.

26.2 Explain why a tranche is a call spread on the pool's loss, identify the strikes, and use Example 26.1 to state the factor by which the pool's and the senior tranche's expected losses each rise when the default probability triples.

26.3 Explain why an agency mortgage pass-through has negative convexity, identify the option and who holds it, and use Example 26.2 to give the effective convexity of the pass-through and of the same bond at a fixed prepayment speed.

26.4 Explain why a mortgage investor hedging to a duration target buys duration as rates fall, and relate this to Section 18.12.

26.5 Name the three sources of credit support in a securitization, explain which is consumed first, and explain why the accumulating-support calculation of Example 26.3 overstates protection in exactly the scenario where protection is needed.

26.6 Explain why mezzanine tranches were pooled and re-tranched rather than sold directly, why doing so did not diversify, and use Example 26.4 to state the range of the "AAA" tranche's expected loss across inter-deal correlations.

26.7 Explain what tail dependence means, why the Gaussian copula's lack of it matters specifically for senior tranches, and why calibrating the correlation to pre-2006 data could not have captured the event that mattered.

26.8 Explain why a AAA tranche's rating is closer to binary than a AAA corporate bond's, and why rating triggers made downgrades destructive rather than merely informative.

26.9 Explain why the May 2005 episode caused losses on both legs of a trade that was supposed to be hedged, and why inferring a correlation from tranche prices and then using it to hedge tranches is circular.

26.10 Describe the originate-to-distribute chain, identify at which point the risk was supposed to stop, and explain how the risk that banks appeared to have distributed returned to their balance sheets.

26.11 Explain why a synthetic CDO multiplies notional without multiplying the underlying default risk, and what this does to interconnection.

26.12 State the sequence in which the crisis unfolded, identify the step at which mortgage losses stopped being the binding problem, and explain why senior tranches largely paid in the end and yet produced enormous losses.

26.13 State four respects in which a CLO differs from a 2007 subprime CDO, say which of them is probably decisive, and use Example 26.6 to explain what is nonetheless worth watching in its collateral.

26.14 Explain what dual recourse means in a covered bond, why it removes the incentive problem, why covered bonds free up no bank capital, and what that implies about lending capacity.

26.15 Explain why the incentive problem in significant risk transfer is inverted relative to 2007,

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and state one respect in which private credit is safer than the pre-crisis structure and one that warrants monitoring.

FURTHER QUESTIONS

26.16 Example 26.1 shows a senior tranche's expected loss rising twenty-eight-fold when the pool's rises threefold. Derive the elasticity of a tranche's expected loss with respect to the pool's default probability, show how it depends on the attachment point, and determine where in the capital structure the elasticity is largest.

26.17 Example 26.4 argues that re-securitization pools the systematic residual. Formalize this: decompose a mezzanine tranche's loss into systematic and idiosyncratic components, show what pooling does to each, and derive the condition under which re-securitization adds diversification rather than concentration.

26.18 Section 26.6 argues that the copula was a bad model used badly, with the second clause carrying most of the weight. Construct the counter-argument that a model with tail dependence would have produced materially different ratings; identify what data would settle the question; and state what your answer implies about the value of model improvement relative to institutional reform.

26.19 Section 26.13 says risk retention addresses the incentive problem but that 5% is small relative to arranging fees. Model the arranger's decision, derive the retention level at which incentives are aligned, and discuss why regulators chose a number below it.

26.20 Section 26.11 states a fair counter-argument that synthetic structures let those who understood the risk be short. Evaluate whether the existence of informed short sellers made the market more or less efficient in this case, and reconcile your answer with the fact that prices did not adjust until the losses arrived.

26.21 Section 26.16 claims that a system loses the ability to see a risk before it loses the ability to bear it. Apply this to private credit: specify what would have to be observed to detect deterioration early, explain why infrequent holder-produced valuations obstruct it, and propose a disclosure regime that would restore observability without forcing mark-to-market on holders who need not sell.

26.22 Compare the 2007–2009 episode with the three episodes of Chapters 18, 22 and 23 along the dimensions of crowding, mechanical hedging, leverage, funding fragility, and observability. Identify which features are common to all four, which are unique to the crisis, and use the comparison to state what a supervisor should monitor that is not specific to any instrument.

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